

ANNA LUKOSE

4th Year Electrical and Computer Engineering student

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Summary

Self-motivated final year Electrical and Computer Engineering student at the University of Cape Town. Largely curious individual seeking high-pressure environments and, therefore, channeling my enthusiasm into practical applications and creative solutions

Achievement

South African Team for High Performance Computing - 2nd Internationally (2025)

With CSIR's CHPC in collaboration with HPE

One of 7 students chosen from hundreds of competitors from various universities to represent South Africa. Placed 2nd in the Virtual International Student Cluster Competition – defeating 23 teams from other countries. Team directly collaborated with and was trained and sponsored by HPE (Hewlett Packard Enterprise).

- Design, configure and optimise high performance computing clusters for speed and efficiency
- Practical exposure and usage of cloud computing principles
- Build (from source) and run various complex applications – including large language models

Work experience

Research: Image-Processing Pipeline for Sinking Marine-Snow Particles (2026)

With MARiS in collaboration with CSIR's Southern Ocean Carbon-Climate Observatory (SOCCO)

Built an end-to-end computer-vision pipeline that segments, tracks and classifies sinking ocean particles from time-lapsed imagery to estimate carbon flux.

- Pre-processing and segmentation to isolate low-contrast, transparent particles with frame-to-frame tracking
- Particle classification through machine learning - complex convolutional neural network training with hyperparameter tuning to create user-specific model

Guest Lecturer (July 2026)

With CSIR's Centre for High Performance Computing (CHPC)

Delivered lectures and trained students from 18 South African universities in high-performance computing fundamentals during the selection round of the CHPC Student Cluster Competition.

Tutor for Embedded Systems I (2026)

With the University of Cape Town

Assisted lecturers in teaching and running practical lab sessions, guiding students through embedded C & microcontroller exercises. Marked coursework and lab assignments, providing feedback to a cohort of over 200 students.

Student Intern (March 2024)

With New Earth Energy, East London

Support experienced engineers with solar design, installation and maintenance while learning the latest advancements in solar technology.

- Design systems based on the requirements of each individual
- Improved communication skills by interacting with clients
- Developed a solid understanding of photovoltaic technology and system efficiency optimisation

Extra Certification

2023 : Huawei ICT Academy – Introduction to Data Communications

Education

2023 – 2026 (*expected*): University of Cape Town: BSc Electrical and Computer Engineering

- 2023 – Present: Academic Dean's List

2018 – 2022: Clarendon High School for Girls, East London

- NSC Results: 94%
- 8th in Eastern Cape (out of 101 588 students) & 1st in Buffalo City Municipality District

Skills

- Java, Python, C++, SQL
- PyTorch, TensorFlow
- Docker, PySpark, Terraform
- Solidworks, Autocad
- Experience with AWS, Microsoft Azure
- Microsoft Power Automate and Power BI
- Git/GitHub, Matlab

Reference:

Name: Miss Mabatho Hashatsi

Company: CHPC

Contact: +(27) 60 526 3070

Relationship: CHPC Mentor